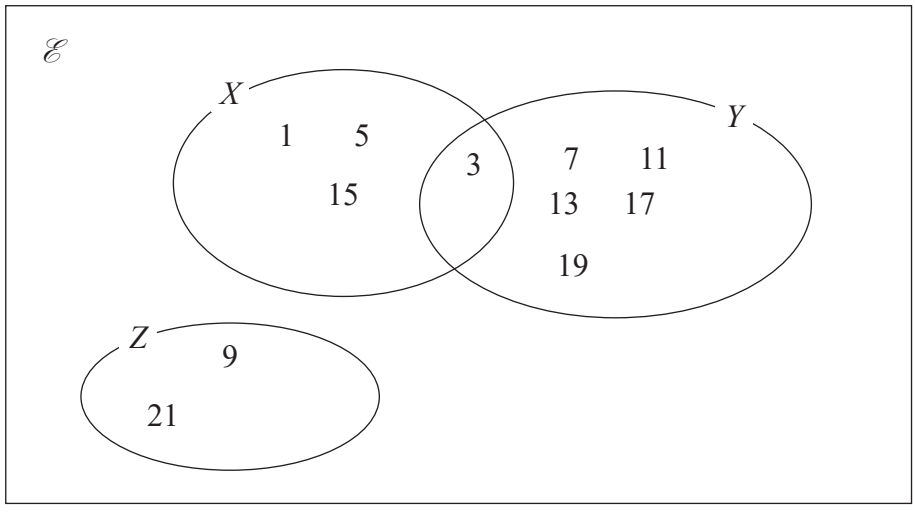
20 Here is a Venn diagram.



(a) List the members of

(i) X

$X = \{1, 3, 5, 15\}$

1 mark
All members inside X.

(1)

(ii) $X \cap Y$

$X \cap Y = \{3\}$

1 mark
In both X and Y.

(1)

(iii) $X \cup Z$

$X \text{ or } Z = \{1, 3, 5, 9, 15, 21\}$

1 mark
In X or Z or both.

(1)

A number is chosen at random from $\mathscr{E}$.

(b) Find the probability that this number is in $Y \cup Z$.

Y or Z: 8 values; universe: 11 values.

1 mark
Count the union and universe.

$\frac{8}{11}$

1 mark
Probability 8 out of 11.

(2)

(Total for Question 20 is 5 marks)